📊 Offline Log Analyzer Tool — User Guide

🌟 Overview

The Offline Log Analyzer Tool helps you inspect and analyze log files to identify:

- Errors and warnings
- System activity
- Internet connections
- USB device usage
- Custom keyword matches

It works completely **offline**, making it ideal for secure environments.

🚀 Features

- 📂 Load and analyze .log files
- 🔍 Detect:
  - Errors
  - Warnings
  - Info logs
- 🔑 Keyword-based search
- 🌐 Internet activity detection
- 🔌 USB connection detection
- 📊 Summary of log statistics
- 🖥️ Simple GUI interface

Getting Started

1. Run the Application

python your_file_name.py

No external dependencies required (uses built-in Python libraries).

How to Use

Step 1: Load Log File

- Click **“Browse Log File”**
- Select a .log file from your system

Step 2: (Optional) Enter Keyword

- Enter any keyword (e.g., failed, login, error)
- Tool will highlight matching entries

Step 3: Analyze Logs

- Click **“Analyze Logs”**

Step 4: View Results

You will see:

- Filtered logs in the main window
- Summary at the bottom:
    - Number of Errors
    - Warnings

- Info entries

🔍 What the Tool Detects

❌ Errors / ⚠️ Warnings / ℹ️ Info

Automatically counts and displays:

Errors: 5, Warnings: 3, Info: 12

🌐 Internet Activity

Detects patterns like:

- connected to internet
- gateway
- dns
- ip address

👉 If found:

System connected to Internet!

🔌 USB Activity

Detects:

- usb device
- removable
- drive letter
- new device

👉 If found:

USB connection detected!

🔑 Keyword Matching

If you enter a keyword:

Example:

login failed

👉 Output:

[Keyword Found] login failed for user admin

📄 Output Display

Logs are shown in format:

2026-03-27 - ERROR: Disk failure detected
[Keyword Found] user login failed
[Internet Detected] IP address assigned

⚠️ Important Notes

- Works best with structured log files
- Assumes timestamp is at beginning of each line
- Large log files may take time to process

🛠 Troubleshooting

❌ File not loading

- Ensure file is .log or text format

❌ No results shown

- Check if file contains:
  - ERROR / WARNING / INFO keywords

❌ Incorrect detection

- Logs must contain recognizable keywords
- Detection is pattern-based (not AI)

🎯 Use Cases

- System log analysis
- Cybersecurity investigations
- Incident response
- Monitoring suspicious activity
- Learning log analysis

🧠 Security Insight

Logs are like a system's memory...

They reveal:

- What happened
- When it happened
- And sometimes — *why it happened*

📝 Final Note

This tool doesn't just read logs...

It listens to the quiet echoes of a system’s past —
connections made, devices plugged in, warnings whispered before failure.